

To Clean or Not to Clean

Senior Design I Ethics Case Presentation

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Project Details

- **Type:** Sustaining Capital
- **Tank Product:** Naphtha
- Tank reported **out of service** for approximately 1.5 years
- **Problem:** Localized weld corrosion on tank shell
- **Solution:** Organize group of contractors to fix welding
- **Budget:** Fixed

EHS Assessment:

- LEL levels inside tank are permissible
- Acceptable ventilation from manways on the side and on top of the tank
- No record of cleaning, possible fire hazard: OSHA States:
“Hot work shall not be performed: On or in equipment or tanks that have contained flammable gas or liquid or combustible liquid or dust-producing material, until a designated person has tested the atmosphere inside the equipment or tanks and determined that it is not hazardous”
– OSHA 1917.152(c) (7) (ii)

Engineering Assessment:

- Previous engineer says “it was cleaned when it was first taken out of service”
- No record of tank cleaning found
- Senior engineers advise that all product would have evaporated after 1.5 years out of service so it is safe
- Supervisor trusts in your ability to carry out repairs ASAP to return tank to service

What will you do?

Choice #1 – Request testing and decide course of action based on results.

Choice #2 – Follow the engineering team's guidance and proceed with repairs

Choice 1) Benefits of Testing

- EHS approved procedure
- Safe environment for the contractors (no fire hazard)
- Avoidance of the extra fees in case of the injuries.
- Keeping the reputation of the company that takes care of its employees



Choice 1) Consequences of Testing (worse case scenario)

- Company **spending extra money** for repairs and cleaning.
- Being considered ‘**not a team player**’ by the engineering team.
- Being **demoted** and not having project leading opportunities in the future.



Choice 2) Benefits of Repairing

- Completes project in **timely manner**
- **Not spending extra** company funds for testing or cleaning.
- **Respected** by the engineering team to take on project management assignments



Choice 2) Consequences of Repairing (worse case scenario)

- Fire hazard in confined space
- Fines/workers comp/lawsuits by employees.
- Fines by OSHA for not following regulations and safety procedures.
- Company gaining reputation of non-safe working environment.
- As an engineer responsible for the project, losing a job and credibility for the future.



Review other choice?

A green rectangular button with rounded corners and a slight gradient, containing the word "Yes" in white text.

Yes

A red rectangular button with rounded corners and a slight gradient, containing the word "No" in white text.

No

AIChE'S Code of Ethics

Scenario # 1



Issues statements or presents information only in an objective and truthful manner.



Act in professional matters for each employer or client as faithful agents or trustees, avoiding conflicts of interest and never breaching confidentiality.

Scenario #2



Hold paramount the safety, health and welfare of the public and protect the environment in performance of their professional duties.



Formally advise their employers or clients if they perceive that a consequence of their duties will adversely affect the present or future health or safety of their colleagues or the public.

**As Chemical Engineers, it is
our responsibility to serve
humanity by making the best
use of our resources and
practice integrity.**

Thank you!